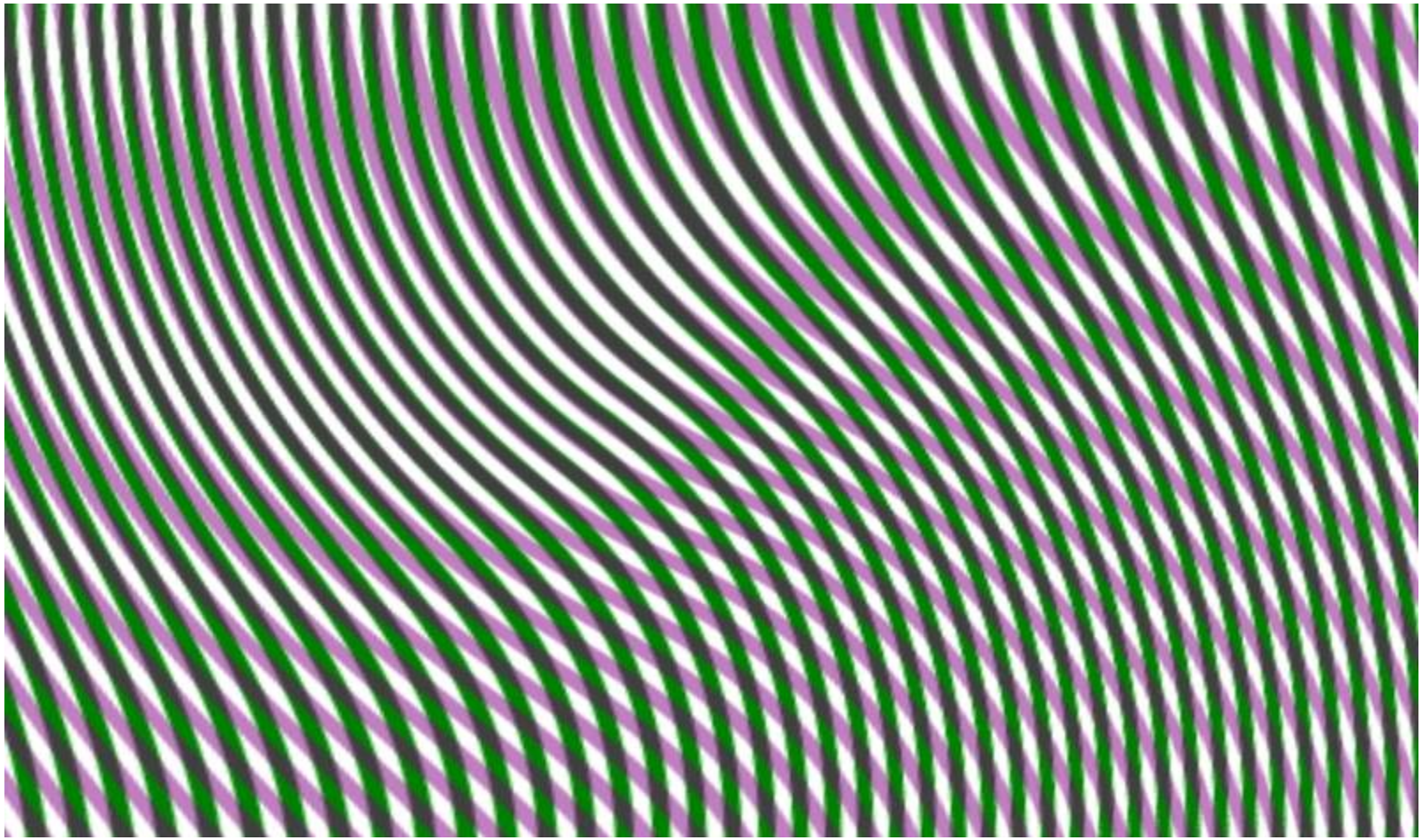
What Does Aliasing Means?

Digital sampling of any signal, whether sound, digital photographs, or other, can result in apparent signals at frequencies well below anything present in the original. Aliasing occurs when a signal is sampled at a less than twice the highest frequency present in the signal. In images, the repetition is in space rather than signals sampled in time for digital audio. If the image data is not properly processed during sampling or reconstruction, the reconstructed image will differ from the original image, and an alias is seen.

An example of spatial aliasing is the moiré pattern one can observe in a poorly pixelized image.

Example of a Moiré pattern



Anti-aliasing*:* Definition

Antialiasing is a technique used in digital imaging to reduce the visual defects that occur when high-resolution images are presented in a lower resolution output devices like the monitor or printer. Aliasing manifests itself as jagged or stair-stepped lines (Also known as jaggies) on edges and objects that should otherwise be smooth.

Before and After of Anti-Aliasing

Aliased polygons (jagged edges)

Anti-aliased polygons

What Does Anti-aliasing Do ?

Anti-aliasing makes these curved or slanting lines smooth again by adding a slight discoloration to the edges of the line or object, causing the jagged edges to blur and melt together. It also removes jagged edges by adding subtle color changes around the lines If the image is zoomed out a bit, the human eye can no longer notice the slight discoloration that antialiasing creates.

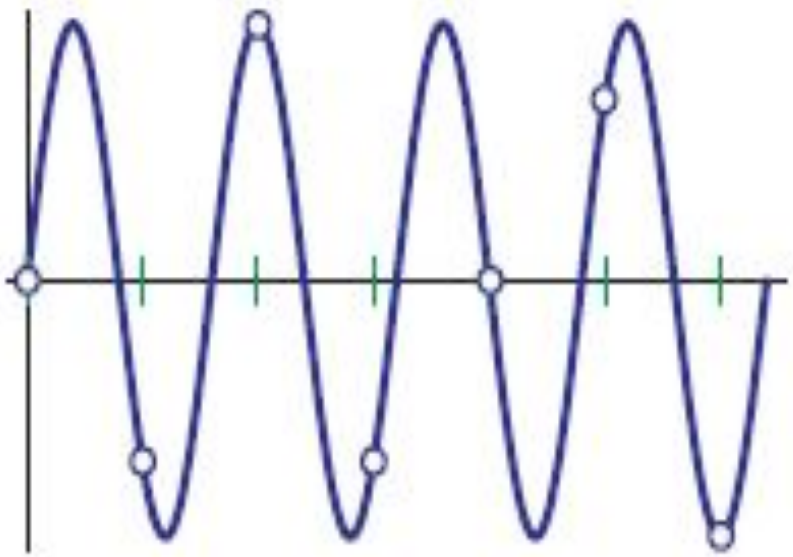
Do We really need Anti-aliasing?

Jaggies appear when an output device does not have a high enough resolution to represent a smooth line correctly. The pixels that make up the screen of the monitor are all shaped in rectangles or squares. Because lighting up only half of one of these square pixels is not possible.

The jagged line effect can be minimized by increasing the resolution of the monitor, making the pixels small enough that the human eye cannot distinguish them individually. This is not a good solution, however, because images are displayed based on their resolution. A single image pixel may take up many monitor pixels, making it impossible for a higher resolution monitor to mask the jagged edges. This is where anti-aliasing is required.

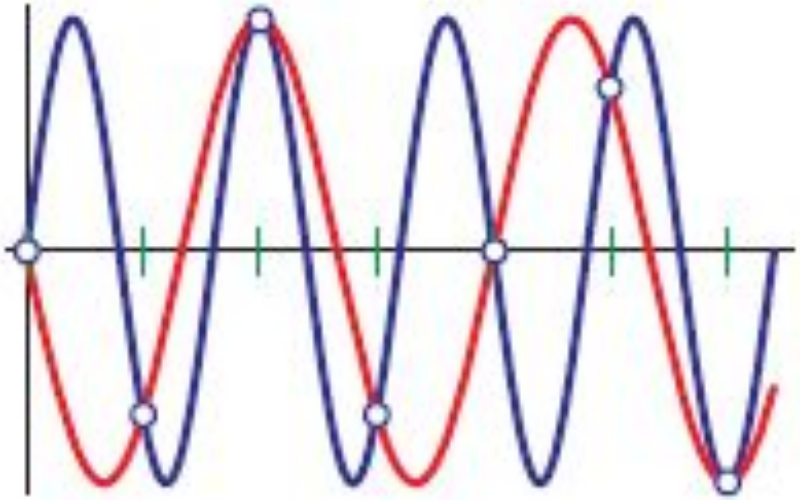
Aliasing in Signal Processing

In signal processing, the term aliasing refers to unwanted artefacts caused by undersampling. Consider the graph of $s(x) = \sin x$, which we sample at $x = 5n\pi/8, n = 0, 1, 2, \ldots$



Aliasing in Signal Processing (cont.)

If the sampled data now used to reconstruct the original signal, an ambiguity is introduced as a lower frequency sine function, $s(x) = -\sin(3x/5)$ could have produced the same data:



In aliasing, high frequency artefacts caused by undersampling mask as a low frequency signal.

ANTIALIASING

This distortion is due to low-frequency sampling and it is called *aliasing*.

To improve the appearance is called *antialiasing* method.

The straight line, circle and ellipse drawing algorithms have to decide which pixel are closest to line.

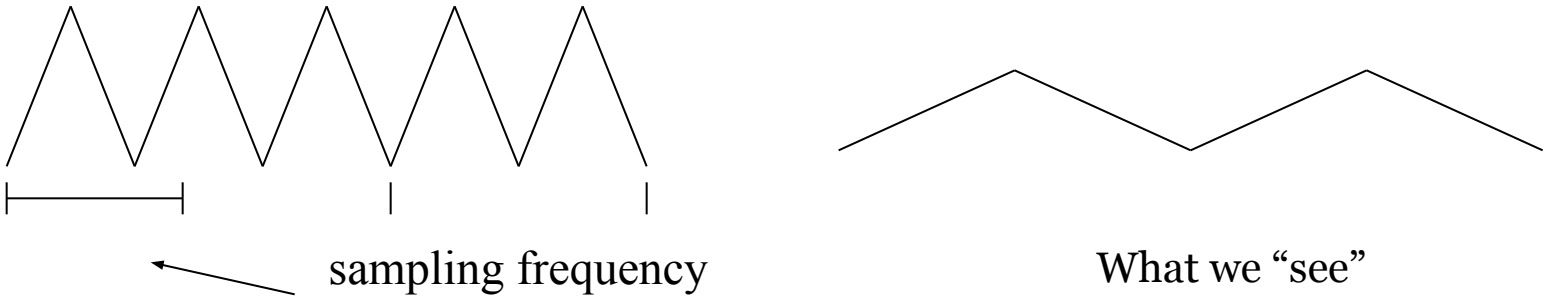
Scan conversion samples a line and choose which of the much smaller number of pixel on the screen to plot to represent a line.

The process of removing one or more of these unwanted artefacts is called antialiasing, or smoothing. Several antialiasing methods are popular in computer graphics:

- Increasing Screen Resolution
- Supersampling
- Area Sampling (Weighted & Unweighted)

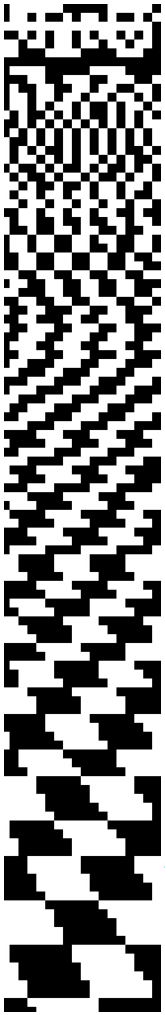
Aliasing

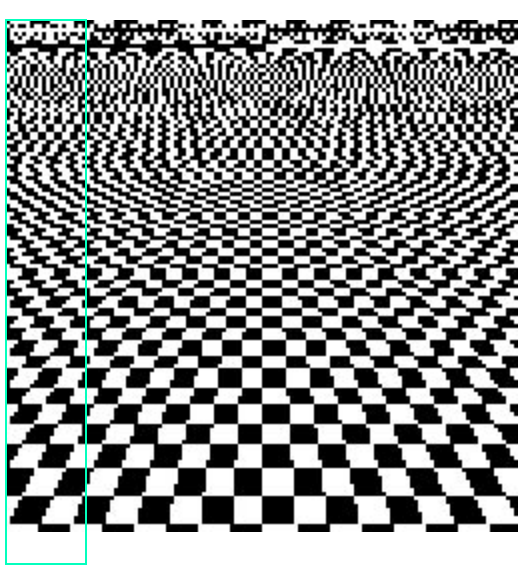
- Distortion of information due to low-frequency sampling



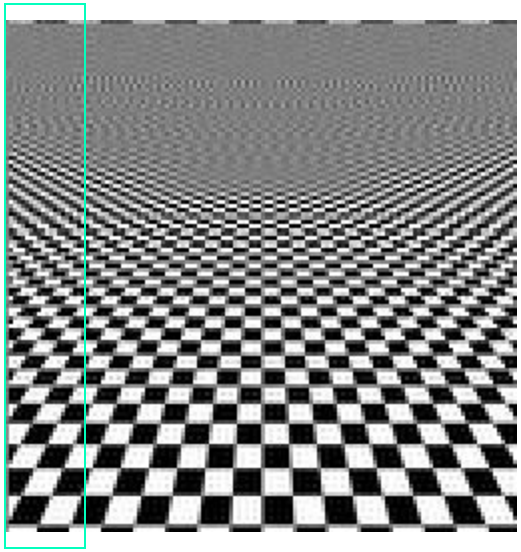
- In raster images – leads to jagged edges with staircase effect. Commonly occurs when a pattern has details that are smaller than a pixel in width
- We can reduce effects by **antialiasing** methods to compensate for undersampling

Antialiasing

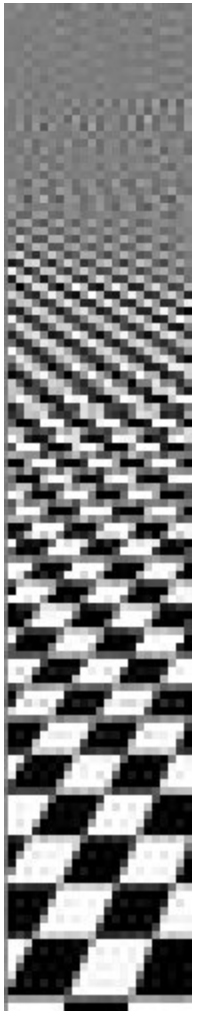




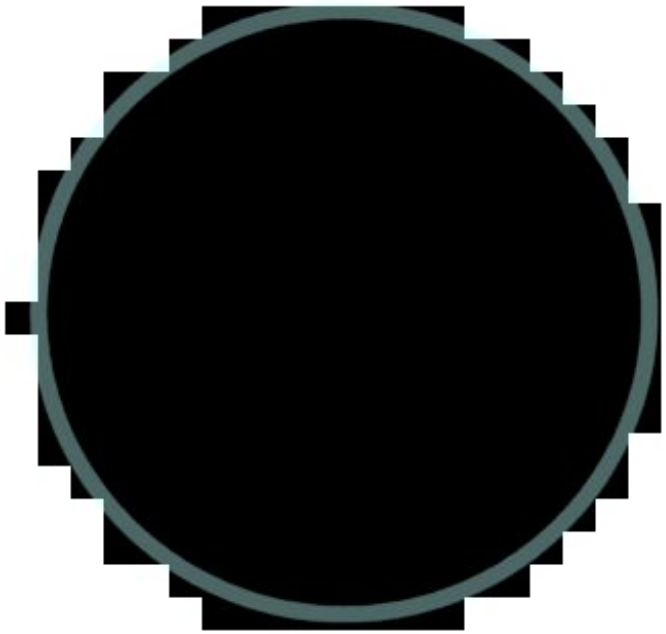
Simple perspective projected texture

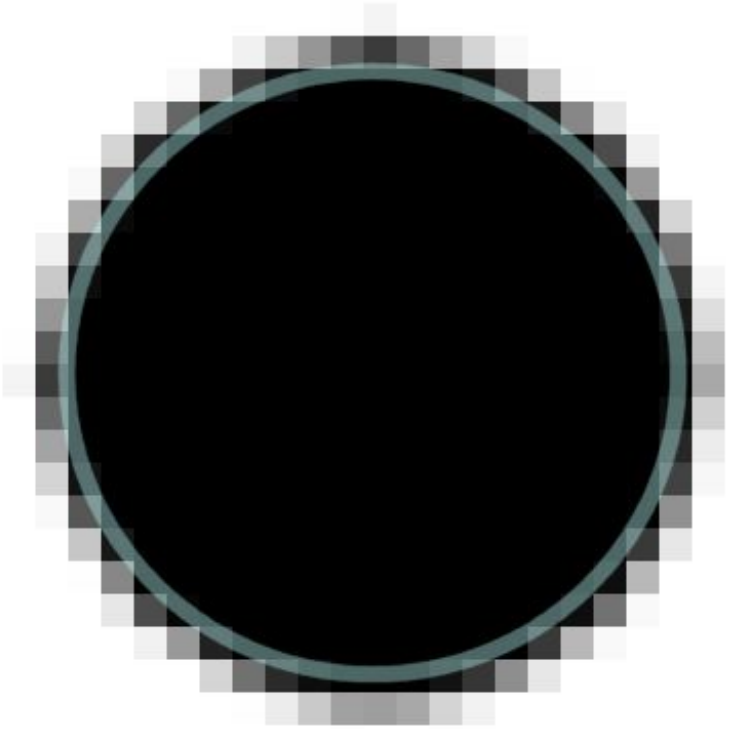


With some antialiasing



Antialiasing





Effectively: BLUR edges to make them look smooth

Nyquist sampling frequency

- Set sampling frequency to at least twice that of the highest frequency occuring in the object to avoid information loss from periodic objects.

$$f_s = 2\, f_{max}$$

Nyquist sampling interval

- Sampling interval should be no more than one-half cycle interval

$$\Delta x_s = \Delta x_{cycle} / 2$$

Where $\Delta x_{cycle} = 1 / f_{max}$

Anti-Aliasing Techniques

Anti-Aliasing techniques were developed to combat the effects of aliasing. There are three main classes of anti-aliasing algorithms :

1.As aliasing problem is due to low resolution, one easy solution is to increase the resolution. This increases the cost of image production.

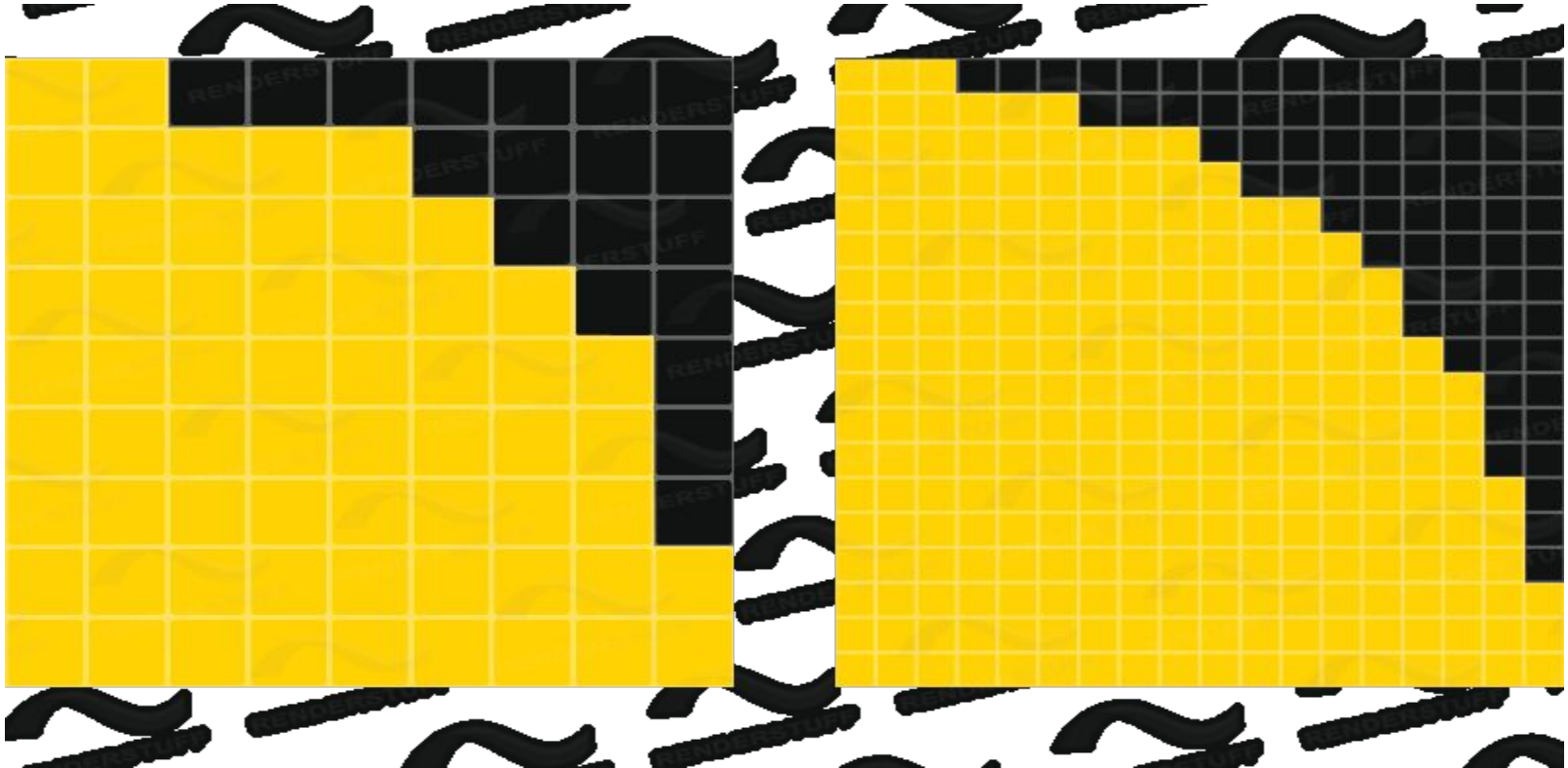
2. The image can be calculated by considering the intensities over a particular region. This is called prefiltering.

3.The image is created at high resolution and then digitally filtered. This method is called supersampling or postfiltering and eliminates high frequencies which are the source of aliases.

4. Unweighted Area Sampling

1. Anti-Aliasing : Increasing Resolution

- Doubling resolution in x and y
  - This method only lessens the problem
- Costs 4 times memory, memory bandwidth and scan conversion time



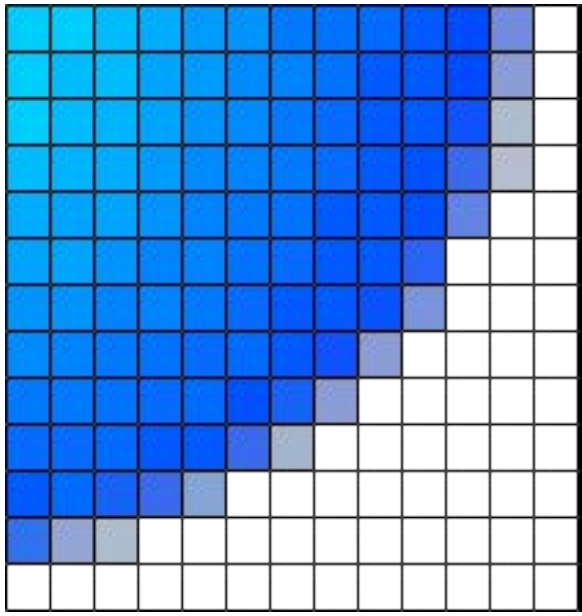
ANTIALIASING

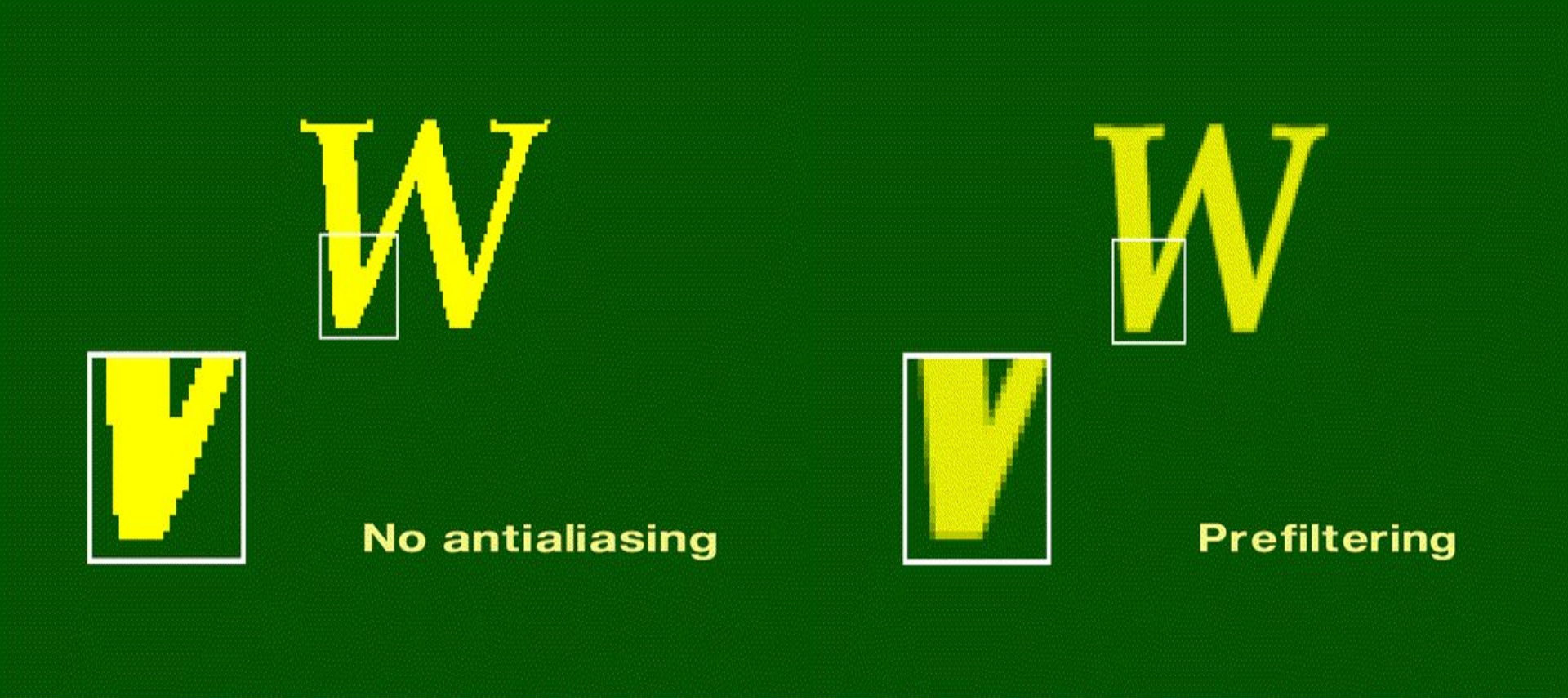
Increasing Screen Resolution

If discontinuities in the model exist, increasing the resolution may make aliasing less noticible, but will not eliminate it, unless the screen resolution exceeds the resolution of the human eye, roughly 0.5′ of arc, or about 600 pixels per inch for an observer 1 to 2 feet from the display.

2. Anti-Aliasing : Prefiltering

Prefiltering methods treat a pixel as an area, and compute pixel color based on the overlap of the scene's objects with a pixel's area. A modification to Bresenham's algorithm was developed by Pitteway and Watkinson. In this algorithm, each pixel is given an intensity depending on the area of overlap of the pixel and the line. So, due to the blurring effect along the line edges, the effect of anti-aliasing is not very prominent, although it still exists. For sampling shapes other than polygons, this can be very computationally intensive.





Original Image

Without antialiasing, the jaggies are harshly evident.

Prefiltered image

.

Along the character's border, the colors are a mixture of the foreground and background colors.

3. Anti-Aliasing : Postfiltering

Supersampling or postfiltering is the process by which aliasing effects in graphics are reduced by increasing the frequency of the sampling grid and then averaging the results down. This process means calculating a virtual image at a higher spatial resolution than the frame store resolution and then averaging down to the final resolution. It is called postfiltering as the filtering is carried out after sampling.

Supersampling is basically a three stage process :

* A continuous image I(x,y) is sampled at n times the frame resolution. This is a virtual image.
** The virtual image is then lowpass filtered.
*** The filtered image is then resampled at the final frame resolution.

Supersampling

1. Construct a virtual higher resolution display by expanding each pixel into $k \times k$ subpixels.
2. Scanconvert the desired graphic into the above.
3. Set the intensity of each actual display pixel to a value that depends upon the n, the number active subpixels. Specifically, this value should be a linear blend between the foreground (F) and background (B) colors:

$$\frac{n}{N}F + \left(1 - \frac{n}{N}\right)B$$

where N represents the maximum number of the $k \times k$ subpixels that can be activated during the scanconversion.

The fraction n/N is called the blend ratio. For RGB models, F and B should be represented by three dimensional vectors.

Supersampling (cont.)

Nonuniform pixel weighting masks can be employed to give the subpixels at the center more weight than those near the boundary.

1	2	1
2	4	2
1	2	1

Let $w_{ij} \geq 0$ denote the weight of the subpixel in row i, column j. The color of each pixel is again computed as a linear blend between the foreground (F) and background (G) colors:

$$\frac{n}{N}F + \left(1 - \frac{n}{N}\right)B$$

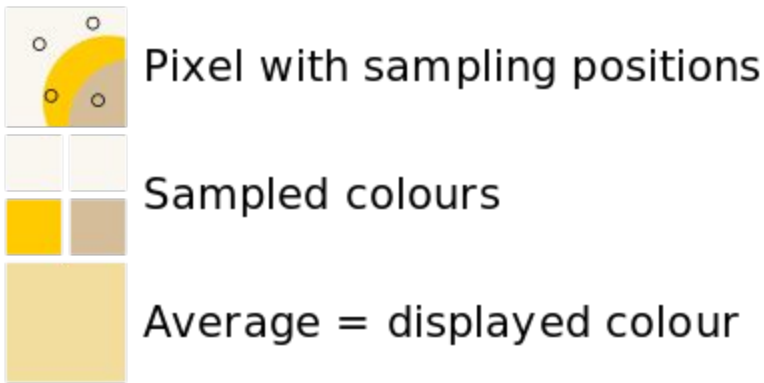
where now, however, n equals the sum of the weights of the active subpixels, and $N = \sum_{i,j} w_{ij}$ equals the total weight of each pixel.

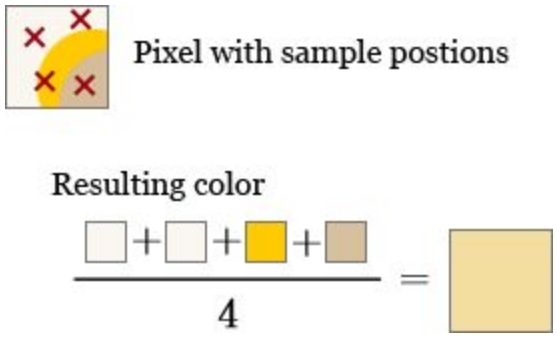
Anti-Aliasing : Postfiltering (Continues)

There are two drawbacks to this method :

There is a technical and economic limit for increasing the resolution of the virtual image.

Since the frequency of images can extend to infinity, it just reduces aliasing by raising the Nyquist limit - shift the effect of the frequency spectrum.





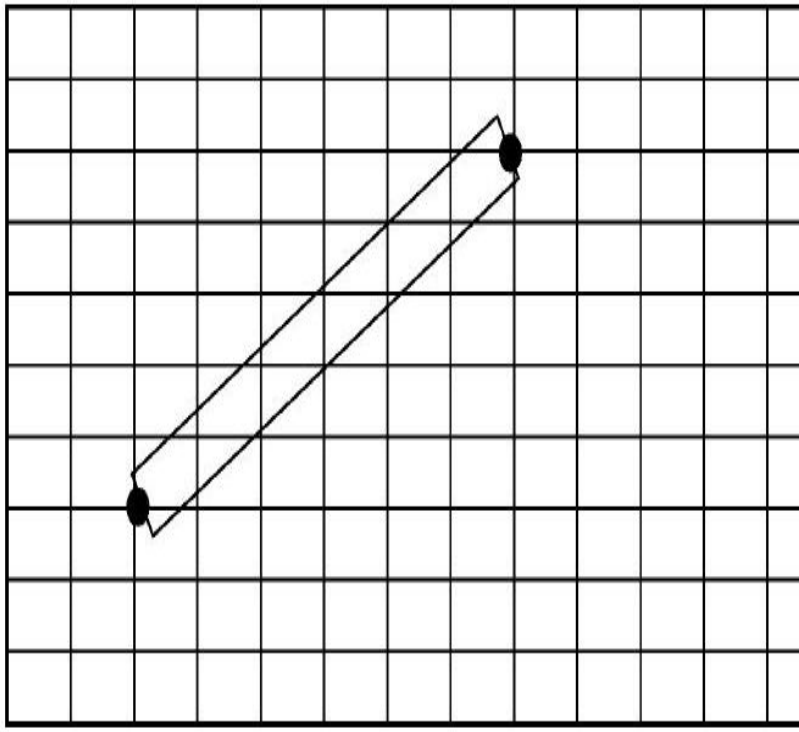
Calculating the end color value

4. Anti-Aliasing : Unweighted Area Sampling

- Drawing a line as a 1-pixel width rectangle.
- For now pixel is unit square centered on

x-y intersection.

- Midpoint algorithm: pick single pixel closet to center line of rectangle. This is a form of point sampling.



Line from point (2,3) to point (8,8)

Anti-Aliasing : Unweighted Area Sampling (Continues)

- Set each pixel's intensity value proportional to its area of overlap covered by primitive
- Note more than one pixel/column for lines of 0<slope<1
- This is a form of unweighted area sampling
  - – the further pixel center is from the line, the less influent it has
  - – only pixels covered by primitive can contribute
  - – only amount of area of overlap matters, regardless of distance of area of overlap from pixel's center

Antialiasing, a method

A polygon will be studied (as an example).

Area sampling (prefiltering): a pixel that is only partly included in the exact polygon, will be given an intensity that is proportional to the extent of the pixel area that is covered by the true polygon

Area sampling

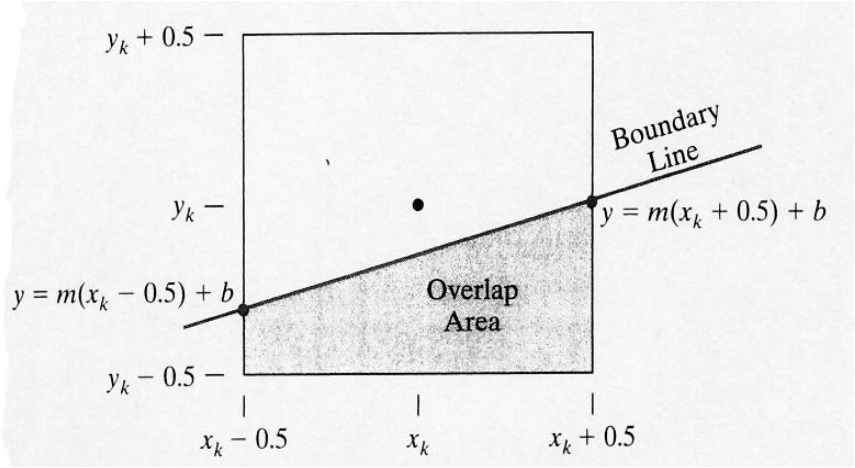
P = polygon intensity

B = background intensity

f = the extent of the pixel area covered by the true polygon

⇒ pixel intensity = $P*f + B*(1 - f)$

Note! Time consuming to calculate *f*



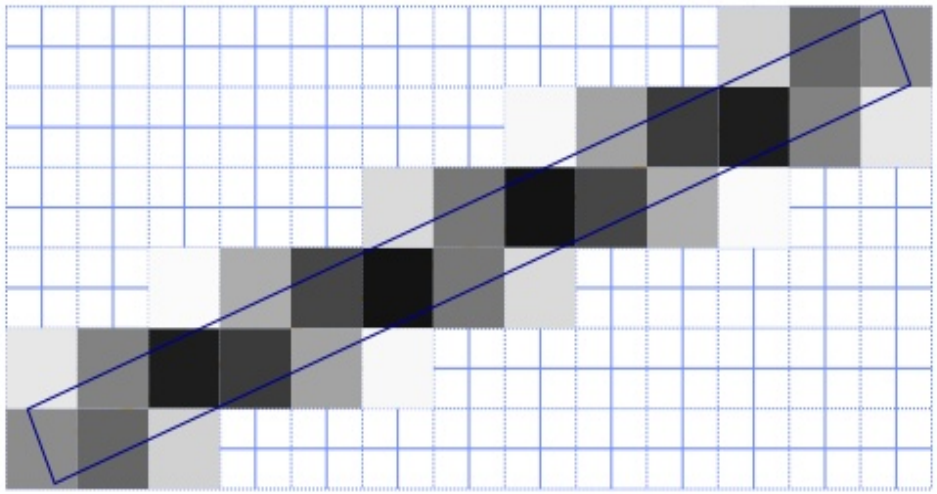
ANTIALIASING

Unweighted Area Sampling (cont.)

Three properties:

1. Pixel intensity decreases as distance from the line center increases.
2. Pixels that do not overlap with the line area are not changed.
3. Equal areas contribute equal intensity, regardless of their relative positions to the line.

Anti-Aliasing : **Unweighted Area**
Sampling



ANTIALIASING

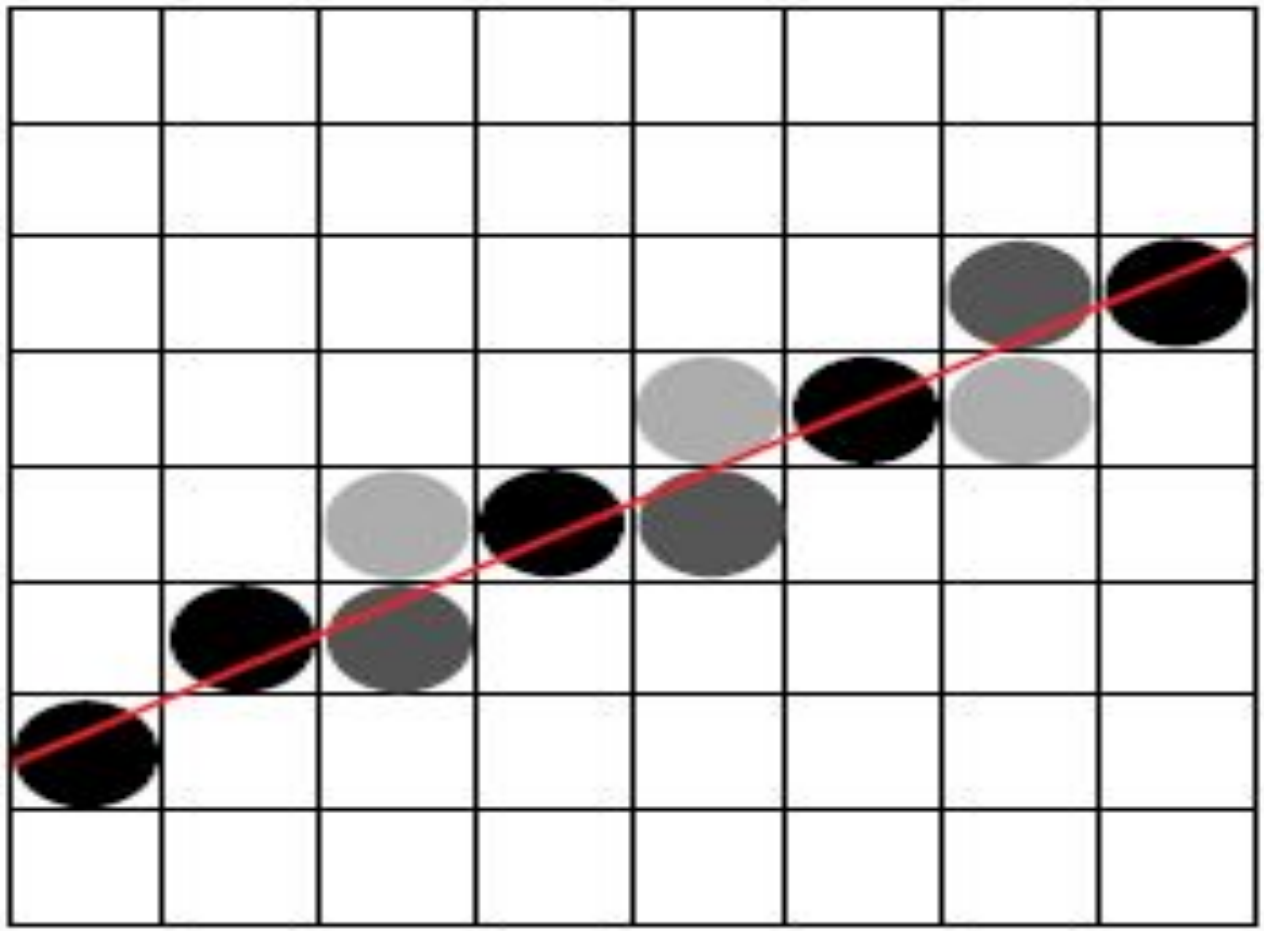
Weighted Area Sampling

Three properties:

1. Pixel intensity decreases as distance from the line center increases.
2. Pixels that do not overlap with the line area are not changed.
3. Equal areas can have unequal contributions: those closer to the line center are weighted more than those that are farther away.

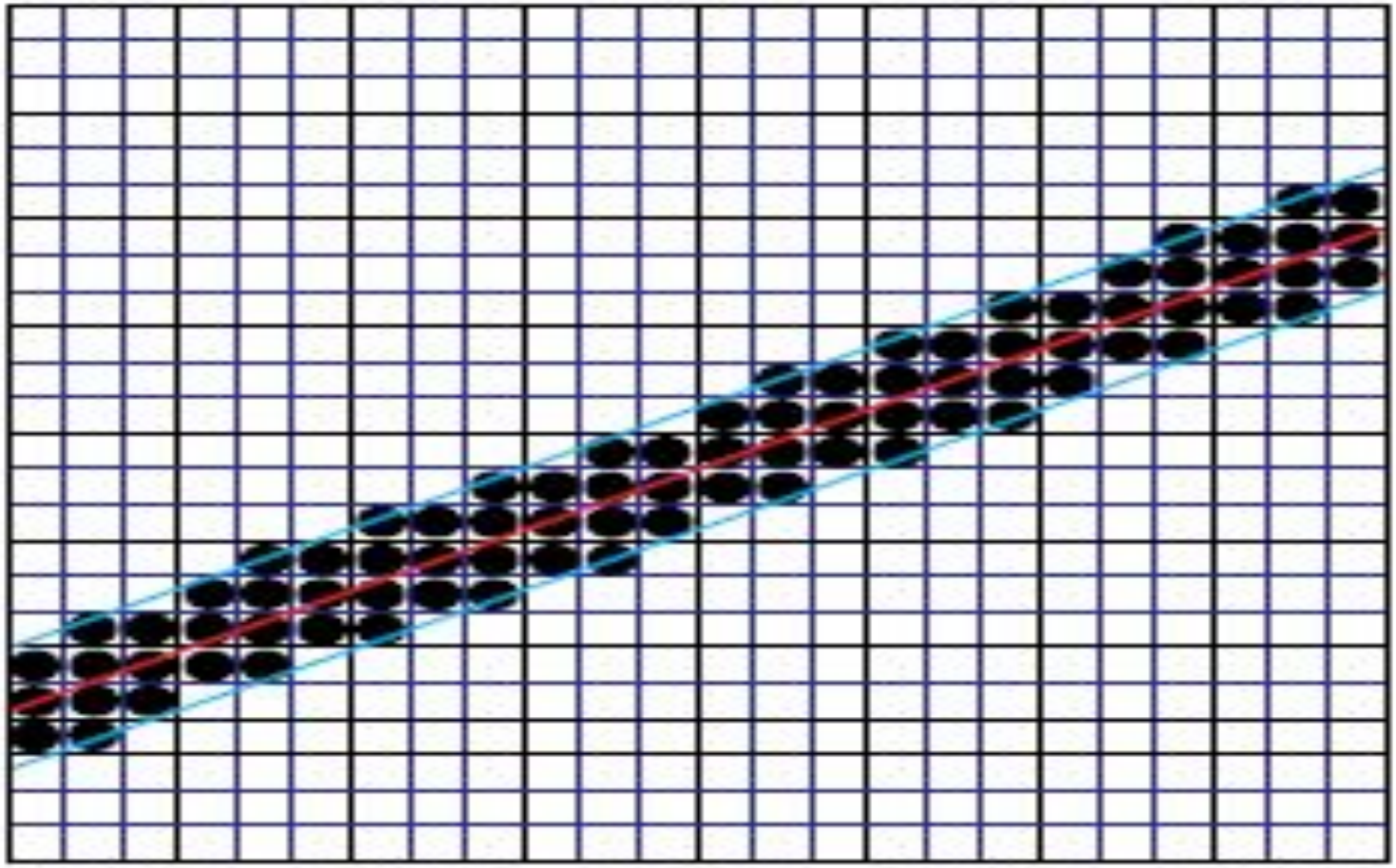
ANTIALIASING

Supersampling with Thin Lines (cont.)



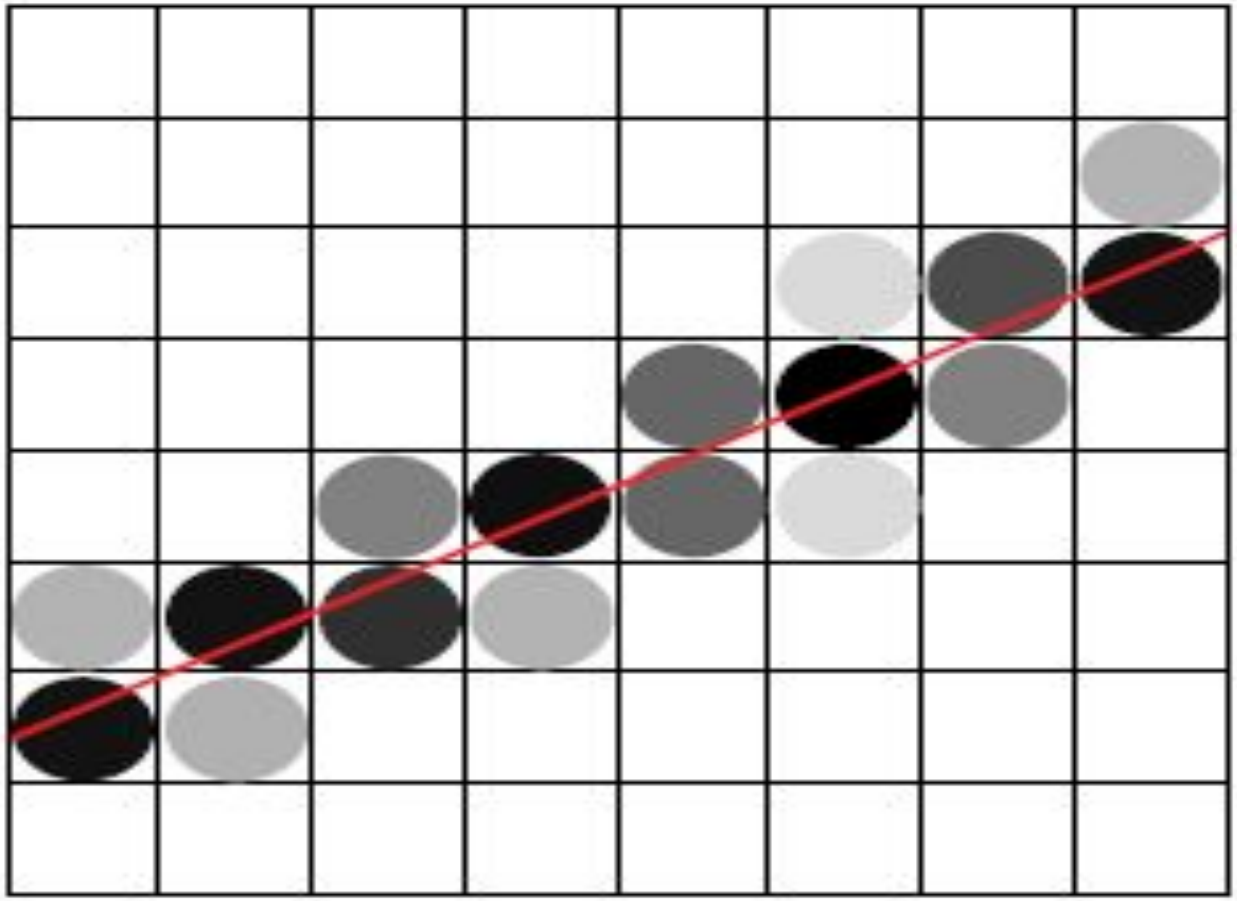
ANTIALIASING

Supersampling with Thick Lines



ANTIALIASING

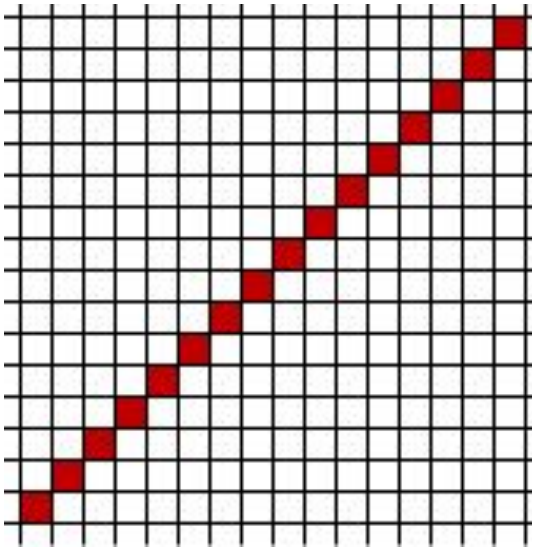
Supersampling with Thick Lines (cont.)

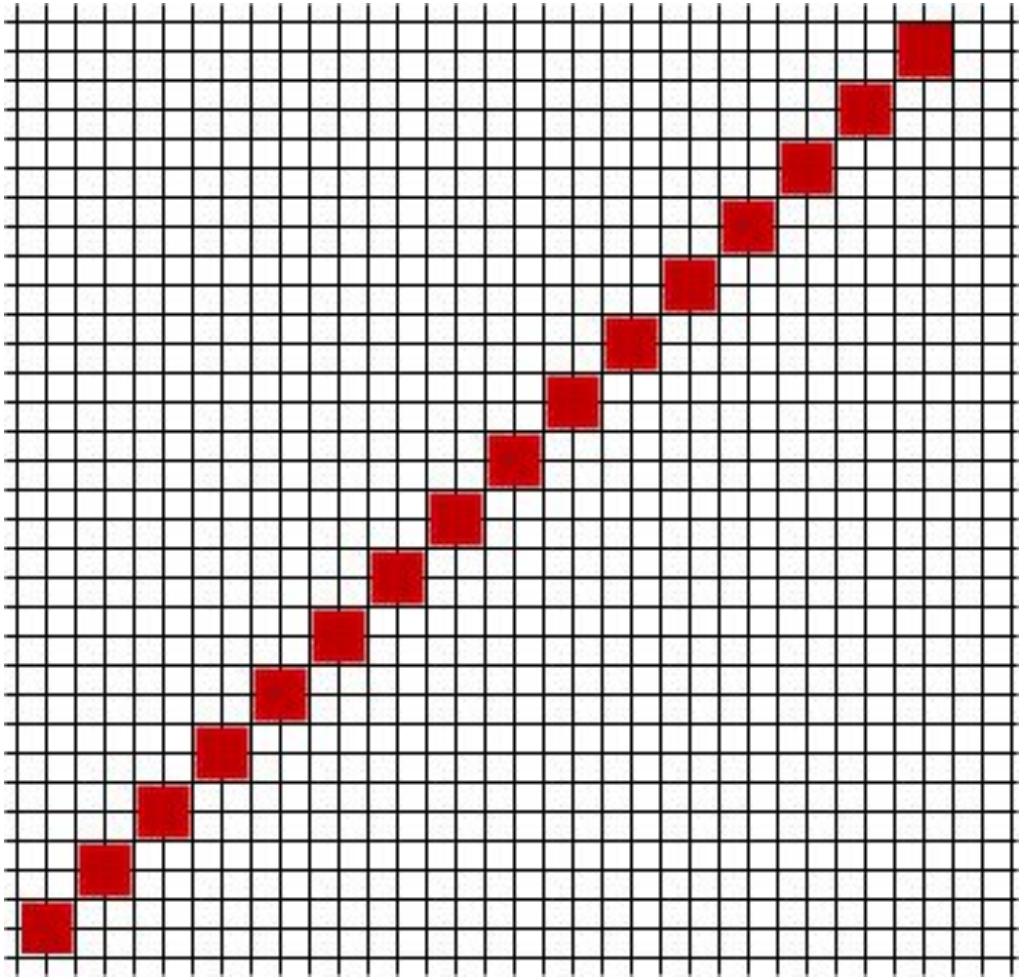


Supersampling Example

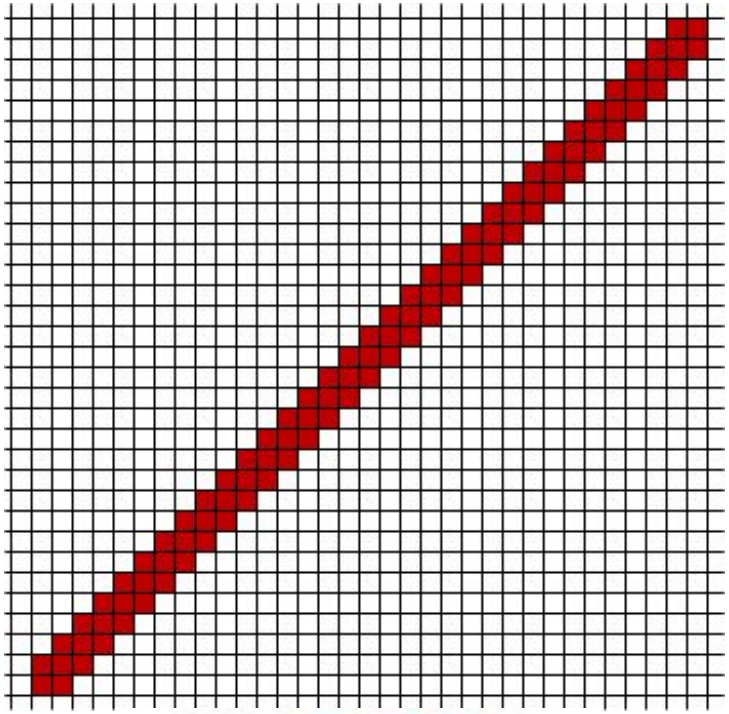
Use twice the available resolution

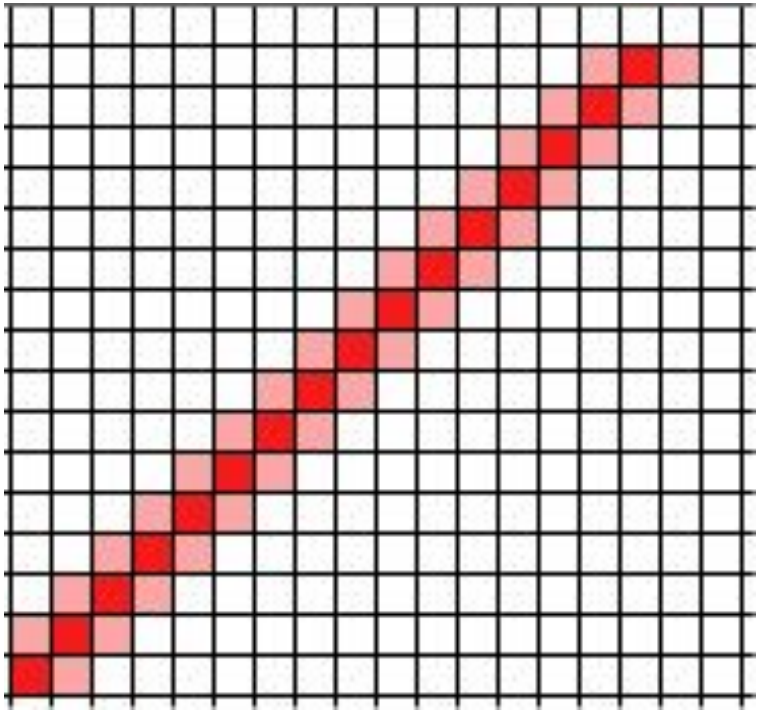
4 pixels (2x2 square) for every 1 in the original

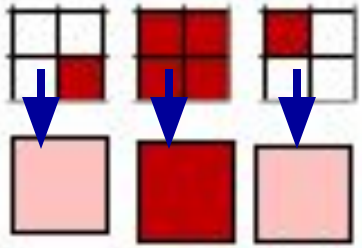


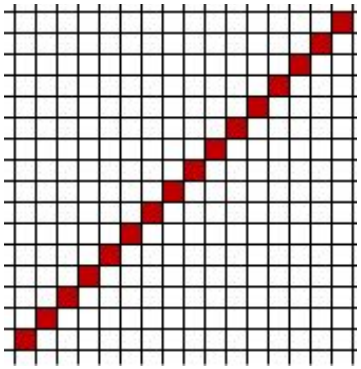


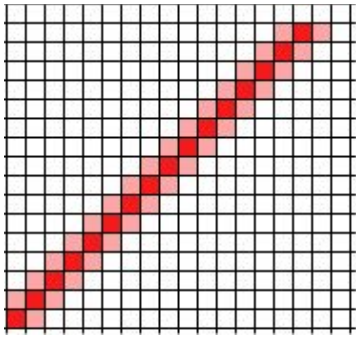
But we don't actually render this hi-res line. Instead we *downsample* it using a "trick"





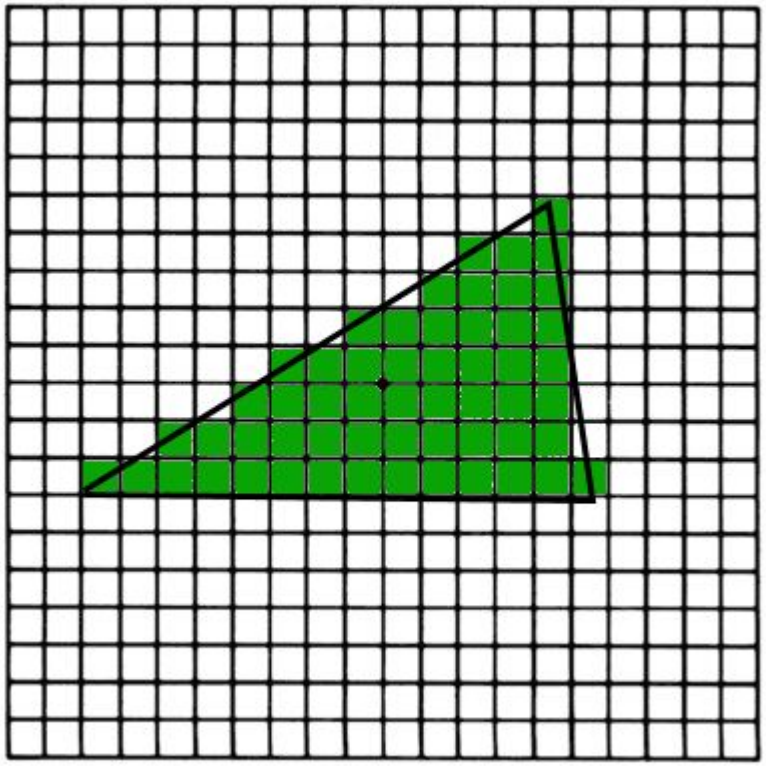


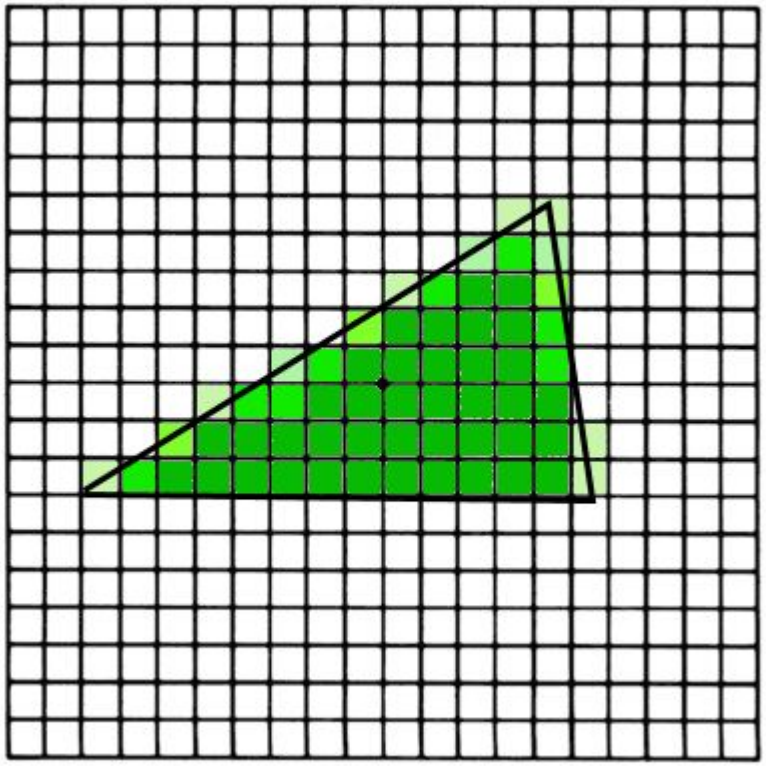


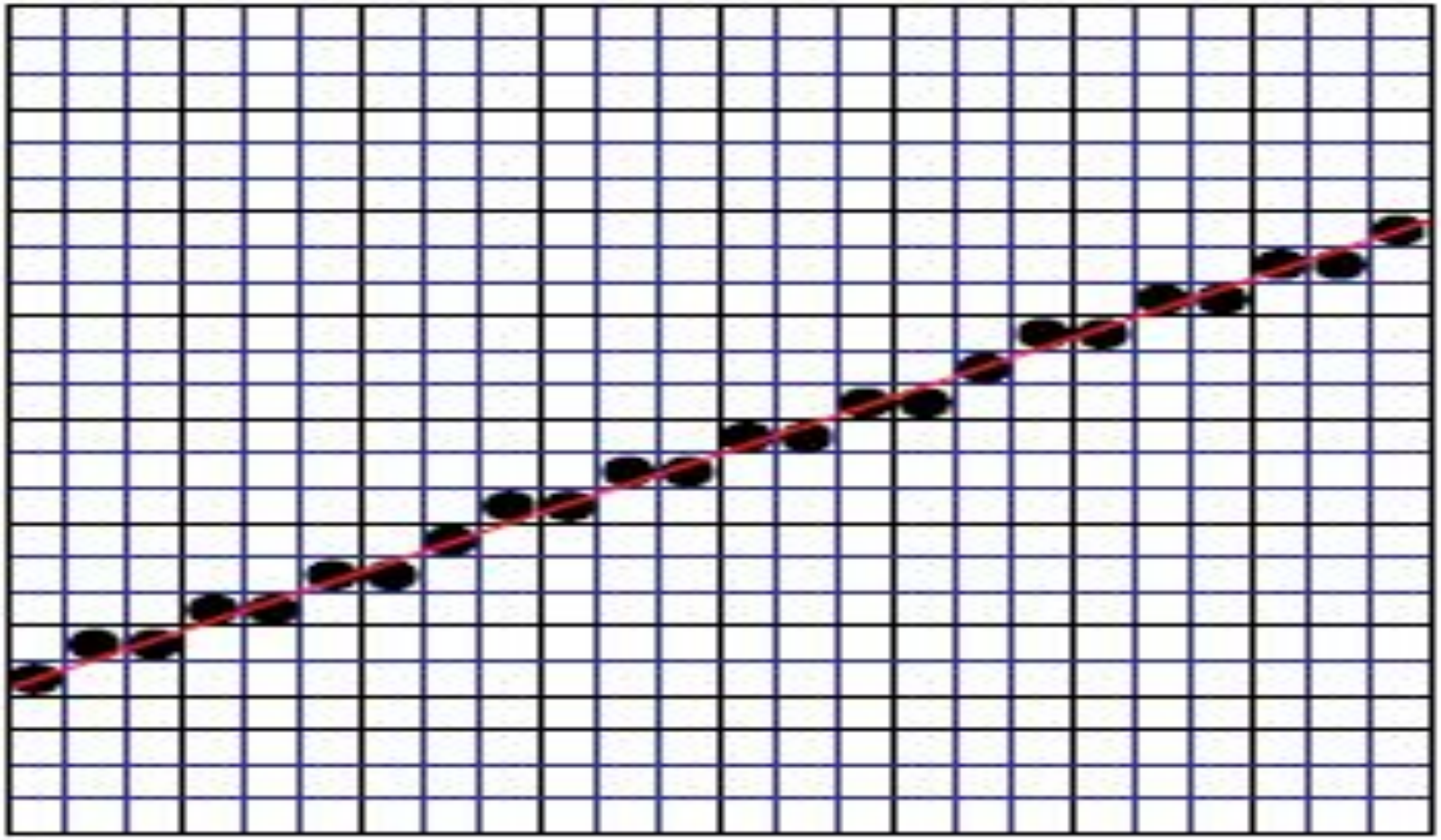


Average out the colours in the 4 pixel squares

Antialiasing by polygon boundary





ANTIALIASING

Supersampling with Thin Lines